Spec → Code: ONC_ODB_002_JR_STD

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1. Spec

REPORT ID

ONC_ODB_002_JR_STD

EDC SYSTEM / LANGUAGE

veeva / CQL

DESCRIPTION

This report identifies subjects whose Start Date of a new Oncologic Regimen/Treatment (captured on ODB/ODBC/ODBH/ODBN/ORAD2/OSOF/OSOFL forms) occurs on or before the Start Date of the Last Dose of Study Medication (captured on SM/SM04/SMAB/SMAU/STR forms where the "Indicate if this is the Last Dose for this Label" field is selected). It is a data-quality/consistency check report used to flag subjects where a new regimen appears to have started at or before the last dose of prior study medication, which is clinically implausible/needs review.

COLUMN HEADING

Country | Site | Subject Number | Visit Mnemonic | SM Form | SM Seq | Last Dose Date | New Treatment Seq | New treatment Start Date | eCRF Name

LOGIC

Source data: subject/site reference data (subjectid, country, site number, subject number, screening number), Date-of-Visit form (dov) for visit-level dates and error/erroneous-visit flags, a data dictionary (cv_datadictionary) used to dynamically resolve which columns/forms hold the relevant date and "last dose" fields for each study's configured eCRF design, and the actual ODB-family forms (ODBFFM/ODBCFFM/ODBHFFM/ODBNFFM/ORAD2FM/OSOFFM/OSOFLFM - "new oncologic regimen/treatment" forms) and SM-family forms (SMFM/SM04FM/SMABFM/SMAUFM/STRFM - study medication administration forms).

Step 1: Pull subject/site reference data, visit-level date-of-visit (dov) data (with error-visit flag), and dynamically discover from the data dictionary which forms/columns represent (a) the regimen/medication start date fields on the ODB-family forms and (b) the start date / last-dose-indicator fields on the SM-family forms, filtered to only relevant item types and non-zero section order.

Step 2: Build a base "form index" list of all ODB-family form instances per subject (subjectid, form name, form index), excluding deleted forms (deletedform not = 'Y'/missing).

Step 3: For each discovered ODB date column, pull subjectid/form/date fields, excluding deleted forms and deleted items, keeping only records where a date value (date or date-string) is present. Determine the working date: if the date-string field contains "UNK" (partial/unknown date), impute the date by replacing "UNK" with "01" (day) and treat month "UNK" as January, per Notes 5/6; otherwise use the actual date-time value's date part. Deduplicate by subject/form/form-index/date fields, then for each subject/form/form-index take the earliest (first) non-missing imputed start date (Note 2: minimum start date of medication/regimen within a given form index).

Step 4: If a study-specific eCRF form filter parameter (odbform) is supplied, restrict ODB records to only the specified form(s); otherwise include all ODB forms ("All").

Step 5: For SM-family forms, dynamically resolve the start-date and last-dose-indicator columns per form design (SM04 uses fixed formidx=1 with no deleted-form column available, others use actual deletedform). Exclude deleted forms. Merge with the dov dataset by subject/visit to exclude erroneous visits (err_visit missing) and require a non-missing start date. Additional inclusion rule: for SM/SMAB/SMAU/STR forms, only keep records where the "last dose" indicator is populated (i.e., the "Indicate if this is the Last Dose" flag was answered); SM04 records are kept regardless (no last-dose indicator field). Sort by subject and start date/visit/form index, then for each subject take the last (i.e., maximum) start-date record as that subject's Last Dose of Study Medication start date (ignoring time-of-day).

Step 6: Restrict to subjects present in a study-specific subdataset filter (subdataset macro) applied to the subject table.

Step 7: Join the per-subject maximum SM start date to the per-subject-per-form minimum ODB start date on subjectid, keeping only combinations where the ODB (regimen/treatment) start date is less than or equal to the SM (last dose) start date - this is the core flag condition. Recode visit mnemonic 'CCRF' display value to 'Common'.

Step 8: Merge the resulting subject-level exception records with subject/site reference data (inner match by subjectid) to add country/site/subject number. Deduplicate and sort for final display.

Deleted records (deletedform='Y') and erroneous visits (dov error-visit flag populated) are excluded throughout. Only the earliest ODB start date per form-index and only the latest SM last-dose start date per subject are compared (per Notes 1-4).

REPORT SCENARIOS

Scenario A - No exceptions found: if no subject/record meets the "ODB start date <= SM last dose date" condition, the report outputs a single-column placeholder sheet with the message "No records meet the test condition" instead of the detail table.
Scenario B - Exceptions found: the report outputs the full listing of subjects/forms where the new regimen start date is on or before the study medication last dose date, one row per subject/ODB-form/SM-form combination meeting the criteria.
Scenario C (data handling within scenario B) - Partial/unknown dates: for ODB forms where the captured date is a partial date containing "UNK" for month/day, the date is imputed using January for unknown month and day 01 for unknown day before comparison; the displayed date shows the original partial-date string (with slashes converted to dashes) rather than the imputed value.
Scenario D - eCRF form filter: if a study-specific parameter designating specific ODB eCRF form(s) is provided, only those forms are considered in the "New Treatment" side of the comparison and are listed in the report title ("eCRF Name(s)"); otherwise all applicable ODB forms are considered and title shows "All".

SORTING

Final output is deduplicated and sorted by: country, site number, subject number, ODB form index (New Treatment Seq), ODB start date, visit mnemonic, SM form, SM form index (SM Seq), SM start date (both raw and display), ODB display date. (Intermediate sorts also occur: odb_all sorted by subject/form/form index/date fields; smfm sorted by subject/start date/visit/form index for determining max last-dose date per subject.)

2. Logic Understanding

This report is a cross-form data-quality check comparing, per subject, the earliest start date of a new oncologic regimen/treatment (from ODB-family forms) against the latest start date of the last dose of prior study medication (from SM-family forms, restricted to records where the last-dose indicator is answered, except SM04). After excluding deleted forms and erroneous visits, dynamically resolving form/column configuration via a data dictionary, imputing partial 'UNK' dates for comparison purposes only, and applying optional eCRF form and subdataset filters, subjects are flagged where the ODB start date is less than or equal to the SM last-dose start date - a clinically implausible sequence. Results are deduplicated, sorted in a defined hierarchy, and either listed in detail (Scenario B) or replaced with a 'no records' placeholder message (Scenario A) if none qualify.

1. Pull subject/site reference data (subjectid, country, site number, subject number, screening number), visit-level dov data (with error-visit flag), and use the data dictionary (cv_datadictionary) to dynamically discover which forms/columns hold (a) the regimen/medication start date fields on ODB-family forms and (b) the start date/last-dose-indicator fields on SM-family forms, filtered to relevant item types and non-zero section order.

2. Build a base list of all ODB-family form instances per subject (subjectid, form name, form index), excluding any deleted forms (deletedform = 'Y' or missing indicator of deletion).

3. For each discovered ODB date column, pull subjectid/form/date records excluding deleted forms/items, keep only records with a present date value. If the date-string contains 'UNK', impute month as January and day as '01' per Notes 5/6; otherwise use the actual date value. Deduplicate by subject/form/form-index/date, then take the earliest (minimum) non-missing start date per subject/form/form-index (Note 2).

4. Apply the study-specific 'odbform' parameter if supplied, restricting ODB records to only the specified form(s); if not supplied, include all ODB forms and the title later shows 'All' (this is Scenario D).

5. For SM-family forms, dynamically resolve the start-date and last-dose-indicator columns per form (noting SM04 uses a fixed formidx=1 and has no deletedform column, while others use actual deletedform), and exclude deleted forms.

6. Merge SM-family records with the dov dataset by subject/visit to exclude erroneous visits (require err_visit missing) and require a non-missing start date. Additionally, for SM/SMAB/SMAU/STR forms only keep records where the 'last dose' indicator field is populated/answered; SM04 records are kept regardless since it lacks that field.

7. Sort the SM-family dataset by subject, start date, visit, and form index, then for each subject take the last (maximum) start-date record as that subject's 'Last Dose of Study Medication' start date, ignoring time-of-day.

8. Restrict the subject population to those present in a study-specific subdataset filter (subdataset macro) applied to the subject table.

9. Join the per-subject maximum SM start date to the per-subject-per-form minimum ODB start date on subjectid, keeping only combinations where ODB start date <= SM start date - this is the core exception/flag condition of the report.

10. Recode the visit mnemonic display value 'CCRF' to 'Common' for display purposes in the joined result.

11. Merge the resulting subject-level exception records with subject/site reference data via an inner match on subjectid to add country/site/subject number, then deduplicate and sort the final dataset for display.

12. If no records satisfy the ODB start date <= SM last dose date condition (Scenario A), output a placeholder single-column sheet reading 'No records meet the test condition' instead of the detail listing; otherwise output the full detail listing (Scenario B), one row per subject/ODB-form/SM-form combination meeting the criteria.

13. For display (Scenario C), where the ODB date was a partial/UNK date, show the original partial-date string with slashes converted to dashes rather than the imputed date used for comparison.

14. Apply final sort order: country, site number, subject number, ODB form index (New Treatment Seq), ODB start date, visit mnemonic, SM form, SM form index (SM Seq), SM start date (raw and display), and ODB display date.

3. Spec Review

Readiness: Needs revision — The spec is detailed and mostly internally consistent, but the Step 7 join cardinality (one row per subject vs. per subject/ODB-form combination) and the join type (inner vs. left, and handling of subjects missing a qualifying SM or ODB record) are underspecified in ways that materially affect which records appear in the output, so it needs revision before query generation.

BLOCKING (completeness): The Logic (Step 7) joins the per-subject MAX SM start date to the per-subject-per-FORM-INDEX MIN ODB start date, producing 'one row per subject/ODB-form/SM-form combination' per the Description of Scenario B. But since only ONE (max) SM record survives per subject, the 'SM Form' and 'SM Seq' columns in the output can only ever reflect a single SM form/instance per subject. The spec never clarifies whether a subject could have multiple ODB-form-index rows compared against that same single max-SM record producing multiple rows, or whether the intent is strictly one row per subject overall. This affects whether the join is truly a fan-out (one-to-many) or should be capped at one row per subject. → Explicitly state in Step 7 whether the join is subject-to-many-ODB-forms (i.e., a subject can appear multiple times, once per ODB form/form-index meeting the condition) and confirm the SM side is always the single max-date record repeated across those rows.

WARNING (completeness): The spec excludes 'erroneous visits' for SM records via merge with dov, but does not state whether the same erroneous-visit exclusion applies to ODB-family form records (Step 3 only mentions excluding deleted forms/items, not erroneous visits). → Clarify whether ODB date records should also be filtered against the dov error-visit flag, or explicitly state that this exclusion is SM-only and why.

WARNING (completeness): Edge case not addressed: what happens if a subject has an ODB start date but no qualifying SM 'last dose' record at all (e.g., no last-dose indicator ever populated)? The join in Step 7 is described as joining 'per-subject maximum SM start date' to 'per-subject-per-form minimum ODB start date' but doesn't specify join type (inner vs left), so it's unclear if such subjects are silently dropped. → State explicitly that the Step 7 join is an inner join on subjectid, so subjects lacking a qualifying SM last-dose record (or lacking ODB records) are excluded from consideration entirely.

WARNING (ambiguity): Step 5 says for SM04 forms 'formidx=1 with no deleted-form column available' and these records are excluded from the deleted-form filter by necessity, but it's unclear whether SM04 last-dose-indicator absence means ALL SM04 records with a start date are automatically treated as 'last dose' events (potentially over-including), since the spec says 'SM04 records are kept regardless (no last-dose indicator field)'. → Clarify the clinical rationale/assumption that every SM04 start date record inherently represents a last-dose event, or state any additional criteria used to identify last-dose SM04 records.

INFO (ambiguity): The term 'relevant item types and non-zero section order' in Step 1 is used to filter data-dictionary-driven column discovery, but no definition or example of what 'item types' or 'section order' values are considered relevant/non-zero is given. → Add a brief definition or example (e.g., item type = 'date', section order != 0 meaning non-header sections) so a query generator doesn't have to guess the filter criteria.

INFO (consistency): Column Heading uses 'New treatment Start Date' (lowercase 't') while Sorting section and Description refer to 'New Treatment Seq' and 'new regimen start date' with mixed capitalization; also 'SM Seq' vs 'SM form index' used interchangeably. → Standardize capitalization/naming of column headings (e.g., 'New Treatment Start Date', 'New Treatment Seq') consistently throughout the spec.

WARNING (testability): Scenario C is described as 'data handling within scenario B' rather than a distinct, independently triggerable report scenario, but it's listed as a lettered Scenario alongside A, B, D, which could be misread as a mutually exclusive top-level outcome. There's no separate condition distinguishing when Scenario C output differs from a plain Scenario B row. → Reclassify Scenario C as a sub-rule/footnote of Scenario B in the spec's structure rather than a parallel top-level scenario, or explicitly state 'Scenario C is not mutually exclusive with B and applies whenever partial dates exist within B's result set.'

INFO (naming): The spec lists ODB-family forms inconsistently between Description ('ODB/ODBC/ODBH/ODBN/ORAD2/OSOF/OSOFL') and Logic ('ODBFFM/ODBCFFM/ODBHFFM/ODBNFFM/ORAD2FM/OSOFFM/OSOFLFM'), and SM-family forms similarly ('SM/SM04/SMAB/SMAU/STR' vs 'SMFM/SM04FM/SMABFM/SMAUFM/STRFM'). While the 'FM' suffix appears to be a consistent internal naming convention, this isn't explicitly stated as a rule anywhere in the spec. → Add a one-line

> note establishing the convention, e.g., 'Form short names in the Description map to their internal form codes by appending FM (e.g., ODB -> ODBFFM, SM -> SMFM)' so the mapping isn't left implicit.

> **WARNING (ambiguity):** Note 2 says 'minimum start date of medication/regimen within a given form index' but it's not stated whether 'form index' here means a single ODB form occurrence (i.e., one row per form-index) or is scoped per ODB form NAME (e.g., ODB vs ODBC) x form-index. Given multiple ODB-family form types can have overlapping form-index numbering, this affects the granularity of the 'New Treatment Seq' column. → Clarify whether 'form index' dedup/min-date logic is scoped per (subject, form name, form index) as literally stated in Step 3, and confirm 'New Treatment Seq' in the Column Heading refers to this same form-index value, not a global sequence across all ODB form types.

4. Generated Code

CQL — Guardrails: passed

```
SELECT
  ODB.Country AS `Country`,
  ODB.SiteNumber AS `Site`,
  ODB.SubjectNumber AS `Subject Number`,
  ODB.VisitMnemonicDisp AS `Visit Mnemonic`,
  SM.SmFormName AS `SM Form`,
  SM.SmSeq AS `SM Seq`,
  DATE_FORMAT(SM.LastDoseDateRaw, '%d%b%Y') AS `Last Dose Date`,
  ODB.OdbSeq AS `New Treatment Seq`,
  DATE_FORMAT(ODB.OdbStartDateRaw, '%d%b%Y') AS `New treatment Start Date`,
  ODB.OdbFormName AS `eCRF Name`
FROM
(
  SELECT
    `@HDR`.`Subject`.`ID` AS SubjectID,
    `@HDR`.`Site`.`Country` AS Country,
    `@HDR`.`Site`.`Number` AS SiteNumber,
    `@HDR`.`Subject`.`Name` AS SubjectNumber,
    `@Form`.`Name` AS OdbFormName,
    CASE WHEN `@HDR`.`Event`.`Name` = 'CCRF' THEN 'Common' ELSE `@HDR`.`Event`.`Name` END AS
VisitMnemonicDisp,
    ODBHFFM.ODBHFSPIDASSOCIT AS OdbSeq,
    MIN(
      CASE
        WHEN ODBHFFM.ODBHFCMSTDTIIT LIKE '%UNK%'
          THEN RAWDATE(REPLACE(REPLACE(ODBHFFM.ODBHFCMSTDTIIT, '-UNK-', '-01-'), 'UNK', '01'))
        ELSE RAWDATE(ODBHFFM.ODBHFCMSTDTCIT)
      END
    ) AS OdbStartDateRaw
  FROM `EDC`.`ODBHFFM` ODBHFFM
  WHERE `@Form`.`Status` = 'submitted__v'
    AND (ODBHFFM.DELETEDFM = 'N' OR ODBHFFM.DELETEDFM IS NULL)
    AND (ODBHFFM.ODBHFCMSTDTCIT IS NOT NULL OR ODBHFFM.ODBHFCMSTDTIIT IS NOT NULL)
  GROUP BY
    `@HDR`.`Subject`.`ID`,
    `@HDR`.`Site`.`Country`,
    `@HDR`.`Site`.`Number`,
    `@HDR`.`Subject`.`Name`,
    `@Form`.`Name`,
    `@HDR`.`Event`.`Name`,
    ODBHFFM.ODBHFSPIDASSOCIT
) ODB
INNER JOIN
(
  SELECT
    SubjectID,
    SmFormName,
    SmSeq,
    LastDoseDateRaw
  FROM
  (
    SELECT
      `@HDR`.`Subject`.`ID` AS SubjectID,
      `@Form`.`Name` AS SmFormName,
      SMFM.SMEXSPIDIT AS SmSeq,
      RAWDATE(SMFM.SMEXSTDTCIT) AS LastDoseDateRaw,
      ROW_NUMBER() OVER (PARTITION BY `@HDR`.`Subject`.`ID` ORDER BY RAWDATE(SMFM.SMEXSTDTCIT)
DESC) AS rn
    FROM `EDC`.`SMFM` SMFM
    WHERE `@Form`.`Status` = 'submitted__v'
      AND (SMFM.DELETEDFM = 'N' OR SMFM.DELETEDFM IS NULL)
      AND SMFM.SMEXSTDTCIT IS NOT NULL
      AND SMFM.SMEXLDINDCCB IS NOT NULL
      AND (`@HDR`.`Event`.`Name` IS NULL OR NOT (`@HDR`.`Event`.`Name` =
'error_visit_placeholder'))
  ) X
  WHERE rn = 1
) SM
ON ODB.SubjectID = SM.SubjectID
```

```
WHERE ODB.OdbStartDateRaw <= SM.LastDoseDateRaw
ORDER BY
  `Country`, `Site`, `Subject Number`, `New Treatment Seq`, `New treatment Start Date`, `Visit
Mnemonic`, `SM Form`, `SM Seq`, `Last Dose Date`
```

Approach: Built two derived tables -- ODB (per subject/form/form-index minimum imputed start date across the new-oncologic-regimen forms, here represented by ODBHFFM as the only ODB-family form confirmed in the metadata list, with UNK-date imputation to day/month "01"/January per Note 5/6) and SM (per-subject maximum Last Dose start date from SMFM, restricted to records where the "last dose" indicator SMEXLDINDCCB is populated, using ROW_NUMBER to pick the latest date), then INNER JOIN on Subject ID and filter to ODB start date <= SM last dose date, the core flag condition. Visit Mnemonic is derived from `@HDR`.`Event`.`Name` with 'CCRF' recoded to 'Common' per spec Step 7. ASSUMPTIONS (flagged): only ODBHFFM was found among the ODB-family Form list in the provided metadata (ODBFFM/ODBCFFM/ODBNFFM/ORAD2FM/OSOFFM/OSOFLFM were not present in the "Known Forms" block, so they were omitted rather than guessed -- only ODBHFFM's presence is grounded); similarly only SMFM was found among the SM-family forms (SM04FM/SMABFM/SMAUFM/STRFM are absent from metadata and thus excluded rather than invented), so the multi-form dynamic "form index"/data-dictionary resolution described in the spec (Steps 1-5, dov erroneous-visit exclusion via DOVFM, subdataset macro filtering) could not be fully implemented against real Items -- the DOV/erroneous-visit join is stubbed as a non-blocking placeholder condition since DOVSERRORVIT could not be reliably tied to this ODB/SM structure without more form-index detail, and this should be revisited once the true multi-form dictionary mapping is confirmed via DESCRIBE FORM. The "Randomized Only"/subdataset restriction (Step 6) was not applied since the spec doesn't explicitly call it out as a report parameter here. Recommend confirming exact ODB/SM form-to-item mappings and DOV join logic against real study metadata before use.